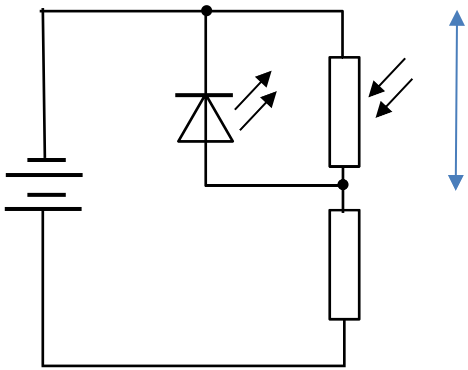
5	
(a)(i)	As the light intensity increases, the resistance of the LDR decreases.
	In Fig. 5.1(a), since the components are in parallel, as the resistance of the LDR varies, the voltmeter reading will remain constant.
	In Fig. 5.1(b), the voltmeter is placed across the fixed resistor so as the light intensity increases by potential divider principle $V = \frac{500}{500 + R_{LDR}} \times 6.0$, the voltmeter reading increases.
(a)(ii)	By the potential divider principle, $V = \frac{500}{500 + R_{LDR}} \times 6.0 = 3.75$
	$R_{LDR} = 300\ \Omega$
(b)(i)	When dark, resistance of LDR increases, p.d. across LDR and LED increase. So the LED turns on. Resistance of LDR decreases. OR When bright, resistance of LDR decreases, p.d. across LDR and LED decrease. So the LED turns off. Resistance of LDR increases.
	The p.d. across LED and LDR decrease, forcing the LED to switch off. OR The p.d. across LED and LDR increase, forcing the LED to switch on.
	When the LED is off, the p.d. across LED and LDR will increase, forcing the LED to switch on. OR

	When the LED is on, the p.d. across LED and LDR will decrease, forcing the LED to switch off.
	 Consider the potential difference across the LED in parallel to LDR
(b)(ii)	A sensible suggestion, e.g. point the LED away from the LDR / increase distance (between LED and LDR) / insert a card between (LED and LDR)
(c)(i)	Horizontal line at 6.0 V.
(c)(ii)	When R = 5.0 Ω, V = 4.0 V. $V = E - Ir$ $r = \dfrac{E - V}{I} = \dfrac{6.0 - 4.0}{(4.0/5.0)}$
	r = 2.5 Ω
6	